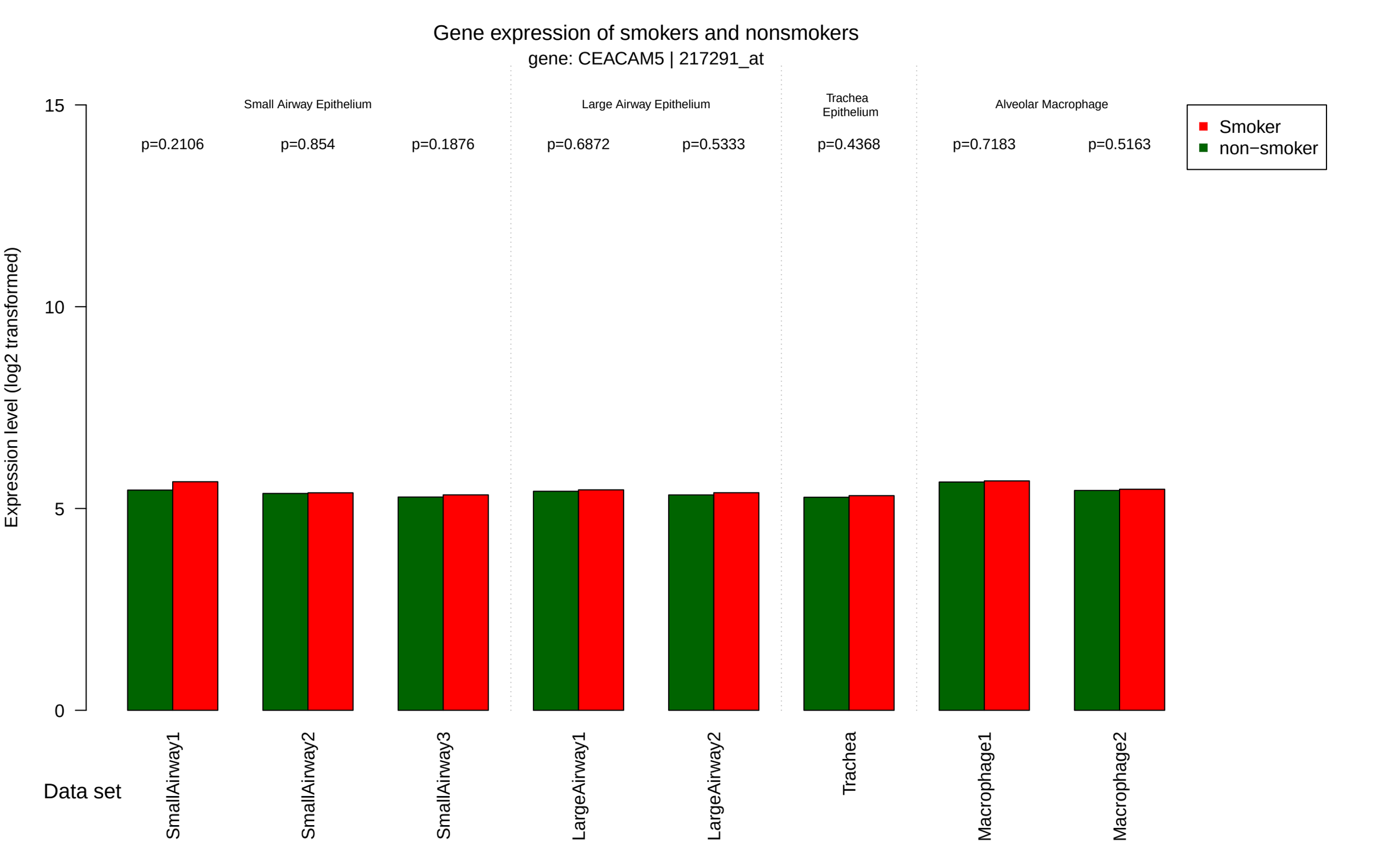
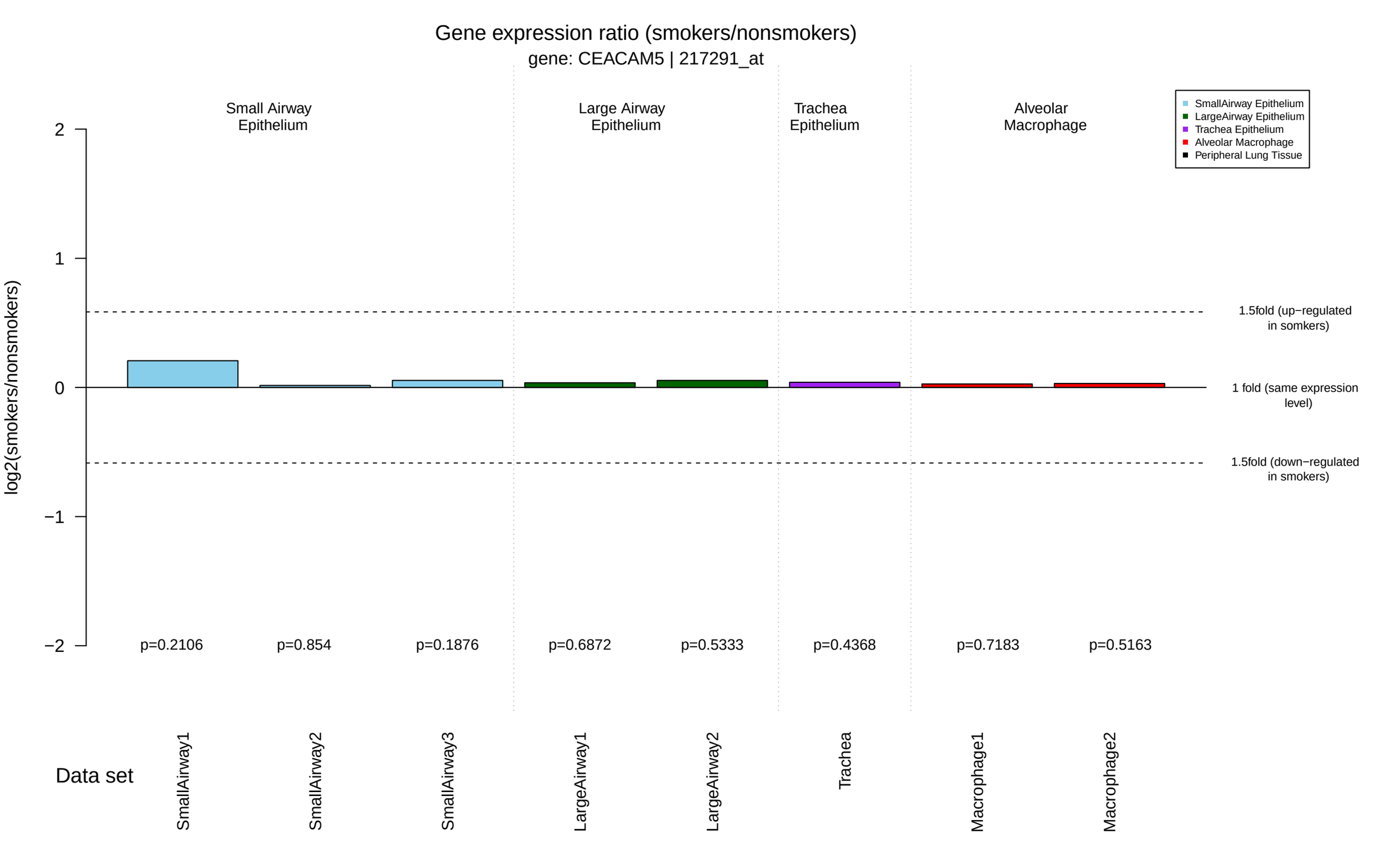
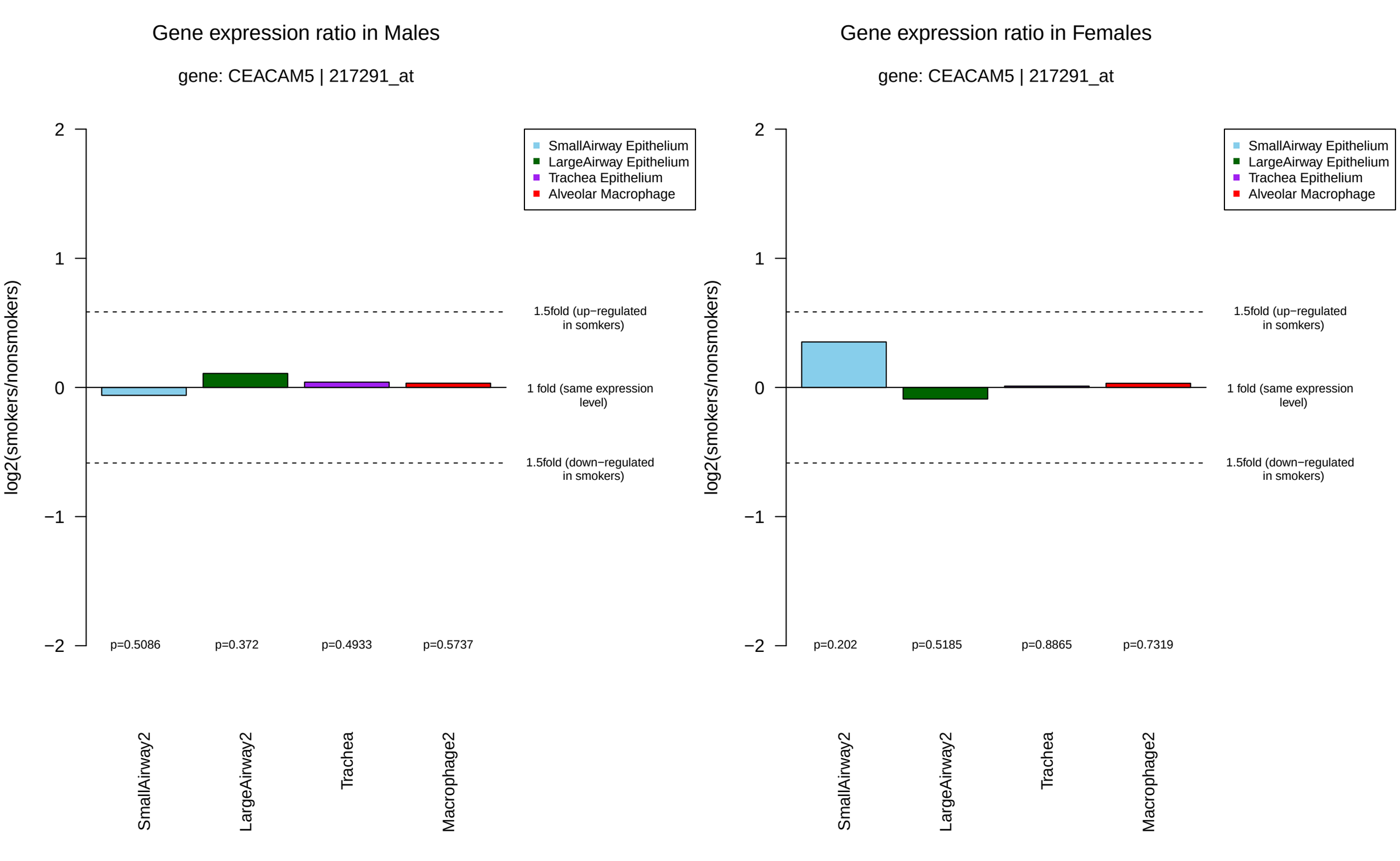
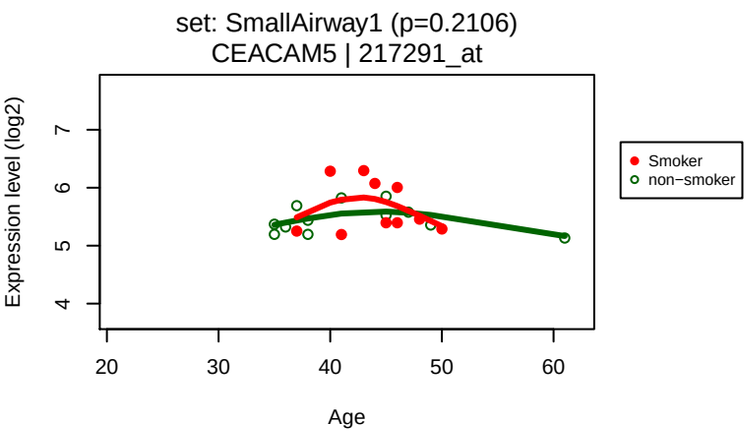
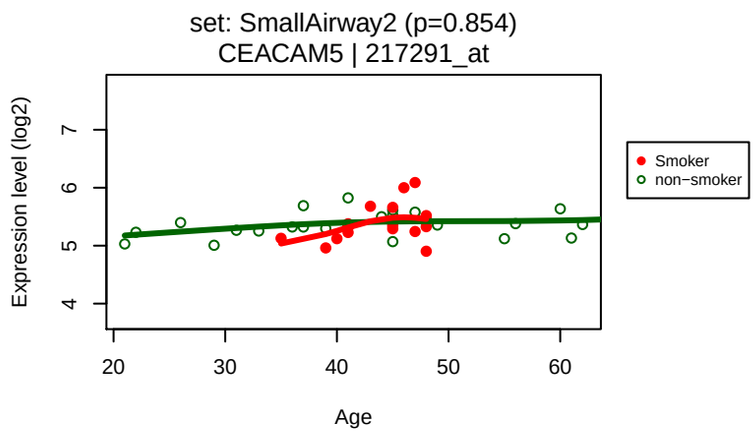
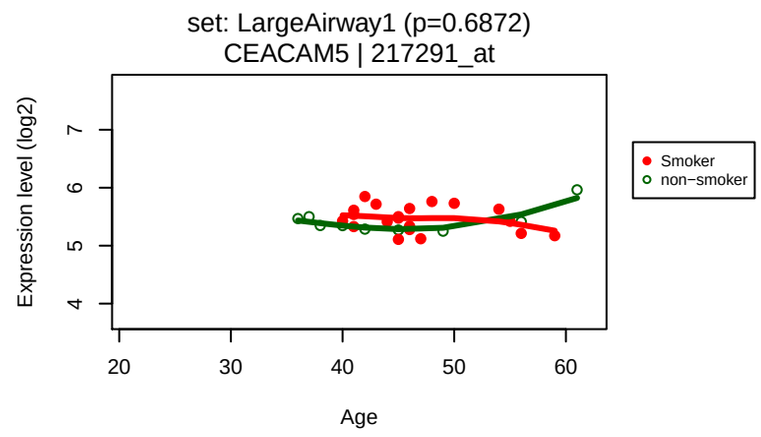

Gene expression of smokers and nonsmokers
gene: CEACAM5 | 217291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2106
p=0.854
p=0.1876
p=0.6872
p=0.5333
p=0.4368
p=0.7183
p=0.5163
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

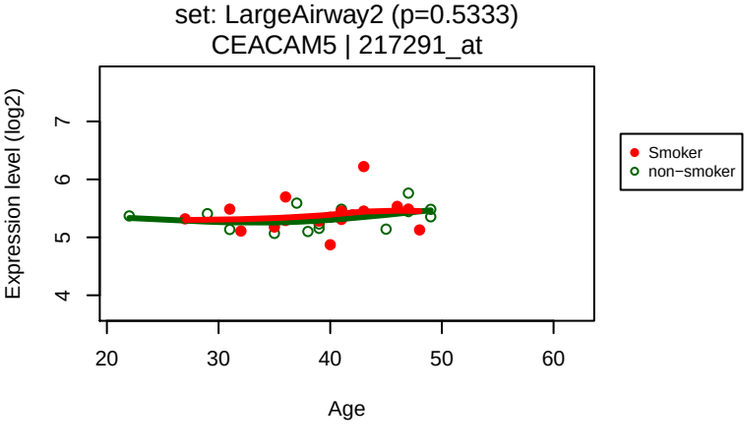

Gene expression ratio (smokers/nonsmokers)
gene: CEACAM5 | 217291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2106
p=0.854
p=0.1876
p=0.6872
p=0.5333
p=0.4368
p=0.7183
p=0.5163
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

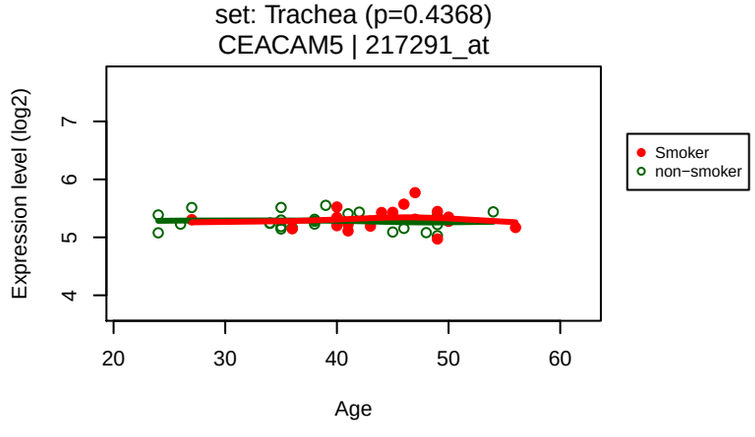

Gene expression ratio in Males
gene: CEACAM5 | 217291_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5086
p=0.372
p=0.4933
p=0.5737
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEACAM5 | 217291_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.202
p=0.5185
p=0.8865
p=0.7319
SmallAirway2
LargeAirway2
Trachea
Macrophage2

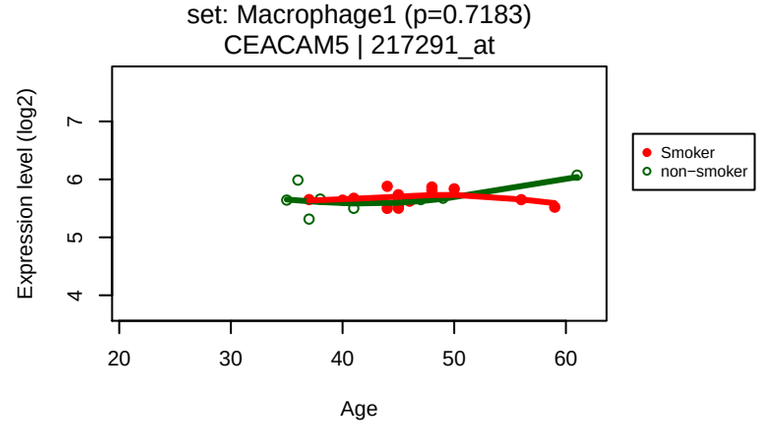

set: SmallAirway1 (p=0.2106)
CEACAM5 | 217291_at
Expression level (log2)
Age
Smoker
non−smoker

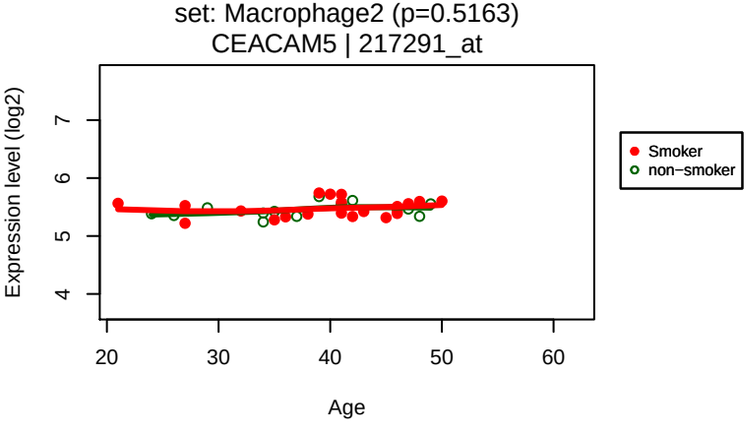

set: SmallAirway2 (p=0.854)
CEACAM5 | 217291_at
Expression level (log2)
Age
Smoker
non−smoker

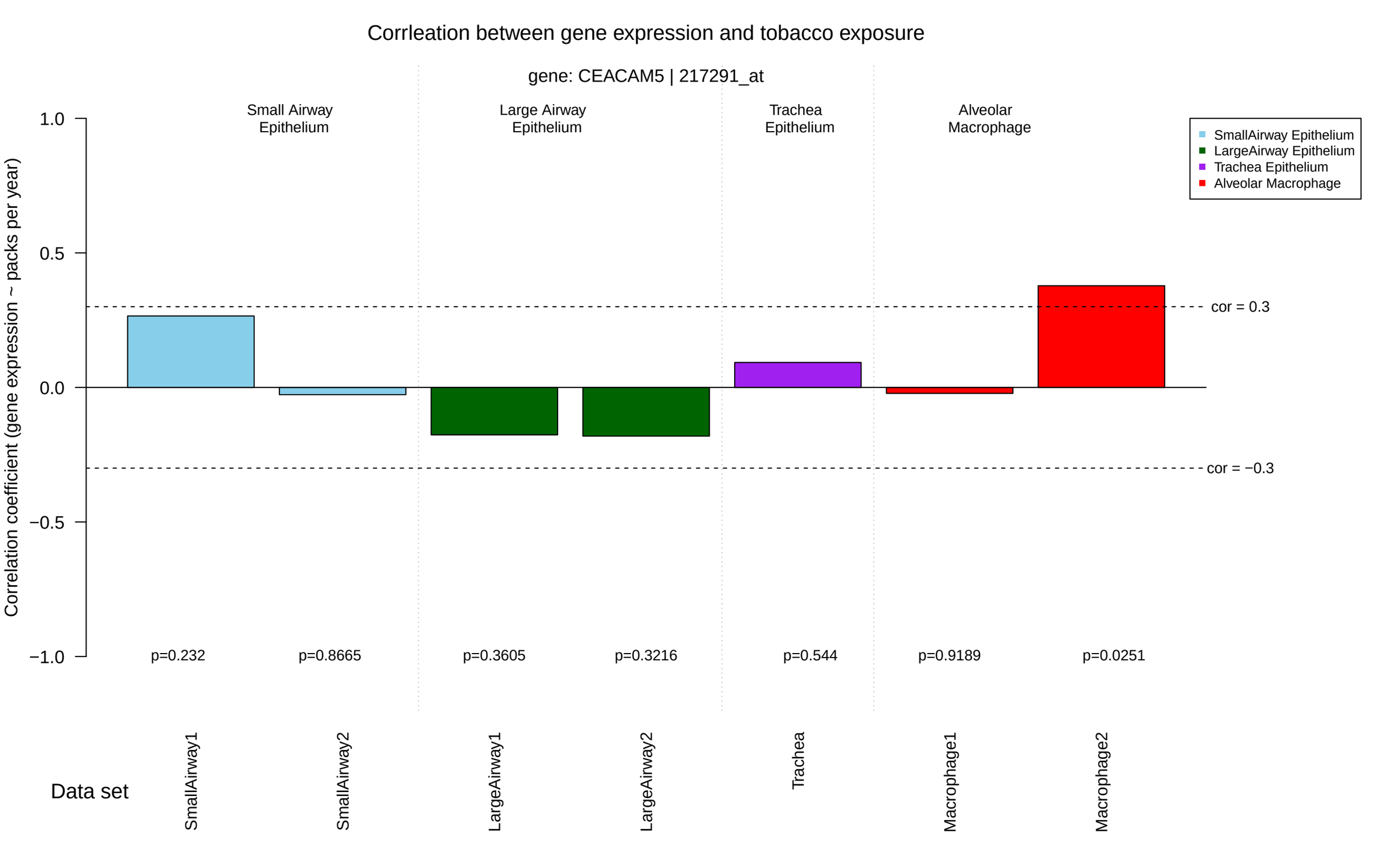

set: LargeAirway1 (p=0.6872)
CEACAM5 | 217291_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5333)
CEACAM5 | 217291_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4368)
CEACAM5 | 217291_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7183)
CEACAM5 | 217291_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5163)
CEACAM5 | 217291_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEACAM5 | 217291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.232
p=0.8665
p=0.3605
p=0.3216
p=0.544
p=0.9189
p=0.0251
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2